

practice_final

 This is a preview of the published version of the quiz

Started: Apr 7 at 11:15am

Quiz Instructions

I am missing a few questions



Question 1 6 pts

The force of attraction that a $-40.0 \mu\text{C}$ point charge exerts on a $+108 \mu\text{C}$ point charge has magnitude 4.00 N . How far apart are these two charges? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: use coulombs law. convert carefully. Isolate the unknown.

and You should also know how to use the inverse square law. IF the force between 2 charges is 10N . What is the force if the distance is $\times 3$? (force is divided by 9). If the distance is divided by 3? (force is multiplied by 9)

☐

3.12 m

☐

1.13 m

☐

2.49 m

☐

3.67 m

☐

2.10 m



Question 2 4 pts

For an electromagnetic wave in free space having an electric field of amplitude E and a magnetic field of amplitude B , the ratio of B/E is equal to

hint: remember E is larger than B in magnitude and c is a big number.

☐

c

☐

$1/c^2$

☐

 square root of (c)

☐
 c^2
☐
 $1/c$
☐

Question 3 4.4 pts

A closed flat loop conductor with radius  2.0 m is located in a changing uniform magnetic field. If the emf induced in the loop is , what is the rate at which the magnetic field strength is changing if the magnetic field is oriented perpendicular to the plane in which the loop lies?

☐

0.080 T/s

☐

2.0 T/s

☐

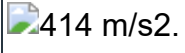
1.0 T/s

☐

0.16 T/s

☐

Question 4 4.4 pts

Two equally charged tiny spheres of mass 1.0 g are placed 2.0 cm apart. When released, they begin to accelerate away from each other at  414 m/s². What is the magnitude of the charge on each sphere,

assuming only that the electric force is present? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: Newton says the forces are always the same (action = reaction) so focus on one charge.

$F = ma$ find F (convert g to kg). Then $F = qE$ with $E = kq/(0.02)^2$. Solve for q .

Here you could write: $F = K q q / (\text{distance})^2$ for the final, you can be given E so $F = qE$

☐

140 nC

☐

75 nC

☐

95 nC

☐

120 nC

☐

Question 5 4.4 pts

A small glass bead has been charged to 1.9 nC. What is the strength of the electric field 2.0 cm from the center of the bead? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

☐

$8.5 \times 10^2 \text{ N/C}$

☐

$8.1 \times 10^{-5} \text{ N/C}$

☐

$4.3 \times 10^4 \text{ N/C}$

☐

$8.5 \times 10^{-7} \text{ N/C}$

☐

Question 6 4.4 pts

A flat square coil of wire with 15 turns and an area of 0.40 m^2 is placed with the plane of its area parallel to a magnetic field of 0.75 T. The coil is flipped so its plane is perpendicular to the magnetic field in a time of 0.050 s. What is the magnitude of the average induced emf in the coil?

hint: $t=0$ there is no flux (no field lines through the area). Then it is maximum. $\text{FLux} = \text{NBA}$.

change in flux = change(NBA)/time = emf

☐

90 V

☐

45 V

☐

6.0 V

☐

36 V

☐

Question 7 4.4 pts

How many electrons are necessary to produce 1.0 C of negative charge? ($e = 1.60 \times 10^{-19} \text{ C}$)

☐

6.0×10^{23}

☐

1.6×10^{19}

☐

6.3×10^{18}

☐

6.3×10^9

☐

1.6×10^9



Question 8 4.4 pts

A straight 1.0-m long wire is carrying a current. The wire is placed perpendicular to a magnetic field of strength 0.20 T. If the wire experiences a force of 0.60 N, what is the current in the wire?

☐

5.0 A

☐

4.0 A

☐

1.0 A

☐

2.0 A

☐

3.0 A



Question 9 4.4 pts

Two point charges each experience a 1-N electrostatic force when they are 2 cm apart. If they are moved to a new separation of 8 cm, what is the magnitude of the electric force on each of them? hint: remember inverse square law. Coulombs force is proportional to the product of the charges but inversely proportional to distance squares.

☐

1/4 N

☐

1/2 N

☐

1/8 N

☐

1/16 N

☐

2 N



Question 10 4.4 pts

An electron moves with a speed of 5.0×10^4 m/s perpendicular to a uniform magnetic field of magnitude 0.20 T. What is the magnitude of the magnetic force on the electron? ($e = 1.60 \times 10^{-19}$ C)

☐

zero

☐ 1.6×10^{-15} N☐ 5×10^{-20} N



$$4.4 \times 10^{-14} \text{ N}$$



$$2.6 \times 10^{-24} \text{ N}$$



Question 11 4.4 pts

How much kinetic energy does a proton gain if it is accelerated, with no friction, through a potential difference of 1.00 V? The proton is 1836 times heavier than an electron, and $e = 1.60 \times 10^{-19} \text{ C}$.
 hint: Potential energy = $q \times \text{voltage}$. If q is in coulombs, energy is in joule. If the charge is in e then energy is in eV. $1 e = 1.6 \times 10^{-19} \text{ J}$



$$1.00 \text{ J}$$



$$1.00 \text{ eV}$$



$$1836 \text{ J}$$



$$1.60 \times 10^{-19} \text{ eV}$$



$$1836 \text{ eV}$$



Question 12 4.4 pts

An electron moving perpendicular to a uniform magnetic field of $3.2 \times 10^{-2} \text{ T}$ moves in a circle of radius 0.40 cm. How fast is this electron moving? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{electron}} = 9.11 \times 10^{-31} \text{ kg}$)

For the final. You can be asked to find the mass of an isotope given r , q , V and B



$$0.80 \times 10^7 \text{ m/s}$$



$$1.9 \times 10^7 \text{ m/s}$$



$$1.9 \times 10^6 \text{ m/s}$$



$$2.2 \times 10^7 \text{ m/s}$$



$$3.0 \times 10^7 \text{ m/s}$$



Question 13 4.4 pts

A tiny particle with charge $+ 5.0 \mu\text{C}$ is initially moving at 55 m/s . It is then accelerated through a potential difference of 500 V . How much kinetic energy does this particle gain during the period of acceleration?

hint: $\text{PE} = \text{KE}$ and $\text{PE} = \text{voltage} \times \text{charge}$.

☐

2500 J

☐

$2.5 \times 10^{-3} \text{ J}$

☐

100 J

☐

$1.0 \times 10^4 \text{ J}$

☐

Question 14 4.4 pts

A 1500-W heater is connected to a 120-V line for 2.0 hours. How much heat energy is produced?

☐

11 MJ

☐

3.0 kJ

☐

1.5 kJ

☐

0.18 MJ

☐

Question 15 4.4 pts

A 200-W light bulb is connected across a 110-V dc power supply. What current will flow through this bulb?

☐

0.90 A

☐

0 A

☐

0.36 A

☐

0.60 A

☐

1.8 A

☐

Question 16 4.4 pts

A 12-V battery is connected across a $100\text{-}\Omega$ resistor. How many electrons flow through the wire in 1.0 min ? ($e = 1.60 \times 10^{-19} \text{ C}$)

Find I first using Ohms law. Then $I = q/t$ so find q . Then $q = N e$. charges are quantized.

☐

2.5×10^{19}

☐

3.5×10^{19}

☐

1.5×10^{19}

☐

4.5×10^{19}

⋮

Question 17 4.4 pts

What current is flowing in a wire if 0.67 C of charge pass a point in the wire in 0.30 s?

hint: Use the definition. $I = Q/t$. number of coulombs flowing per sesonc.

☐

0.30 A

☐

0.20 A

☐

0.67 A

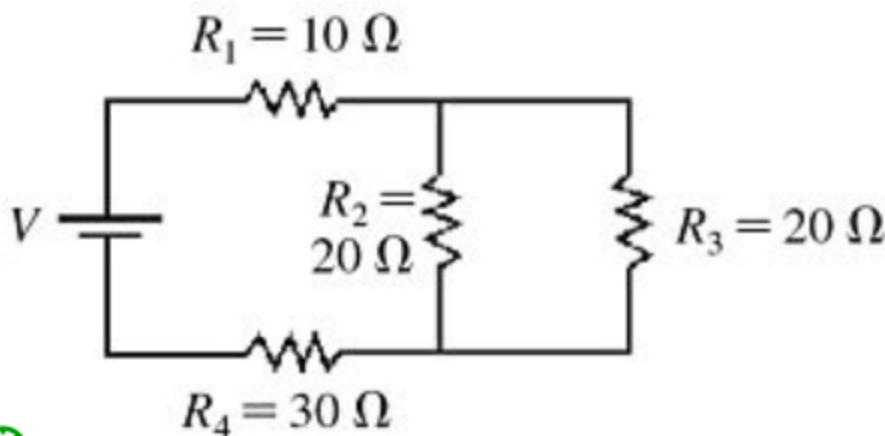
☐

2.2 A

⋮

Question 18 4.4 pts

If $V = 20$ V and the battery is ideal, what is the current through R_3 in the figure?



hint: find the equivalent resistance of the 2 branches first (20 and 20). You get a circuit in series.

power supply - V_1 - V_2 - V_3 .

V_1 is the voltage drop across 10, V_2 is the voltage drop across the middle resistance and V_2 is the voltage drop across 30 ohms. Then find the total equivalent resistance from the 3 resistances. Compute the main current (drained from power supply). Go back one step. Find V_1 , V_2 , V_3 since you have I_{main} .

One step back you have I_2 and I_3 from Ohms law. TADA! Physics rocks!



0.20 A



0.050 A



1.0 A



4.0 A



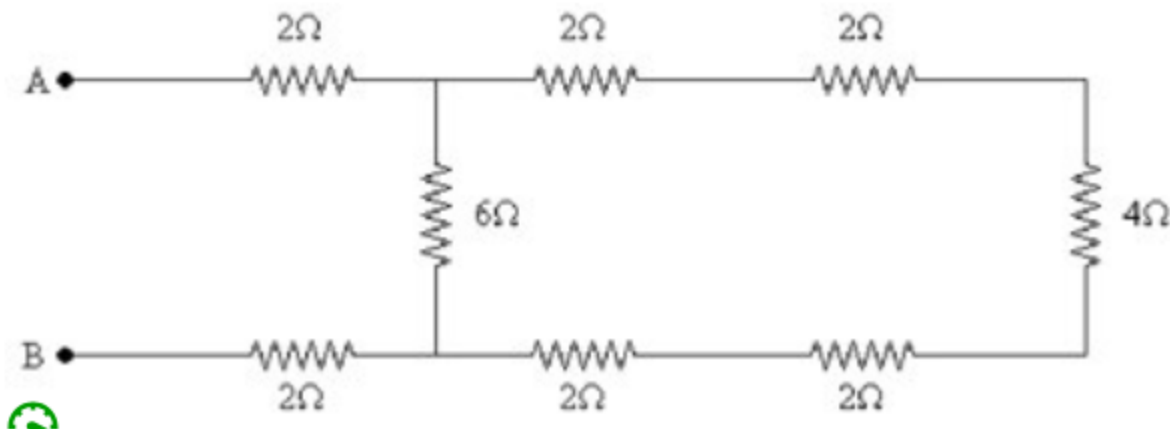
Question 19 4.4 pts

A doubly charged ion with speed 6.9×10^6 m/s enters a uniform 0.80-T magnetic field, traveling perpendicular to the field. Once in the field, it moves in a circular arc of radius 30 cm. What is the mass of this ion? ($e = 1.60 \times 10^{-19}$ C)

 8.2×10^{-27} kg 6.7×10^{-27} kg 11×10^{-27} kg 3.3×10^{-27} kg

Question 20 4.4 pts

A number of resistors are connected across points A and B as shown in the figure. What is the equivalent resistance between points A and B?



hint: step1: 2,,2,4,2,2 are in series. find Req1. step2: Req1 and 6 are in parallel. Find Req2. 2, Req2 and 2 are in series. Easy !

12 Ω

4 Ω 8 Ω 10 Ω 6 Ω 

Question 21 6 pts

A light bulb produces 10W/m² at a distance of 5 m (so 5 m away from the source, 10J of energy flows through a square 1m x 1m every second).

hint: Brightness = Luminosity / (4 π R²) . Luminosity is the power of the source. brightness is the power at a distance R per unit area. Find Luminosity (W)



about 3000W



about 30W



about 300W



about 30,000W



Question 22 4.4 pts

At a certain instant in time, the electric field of an electromagnetic wave in free space points in the -z direction, and the magnetic field points in the +y direction. In what direction is this wave traveling?

hint: right hand rule. (V, E, B). Very Educated Beaver? (Velocity, Electric field, magnetic field)



+x direction



+z direction



-z direction



-x direction



+y direction



Question 23 4.4 pts

Which one of the following lists gives the correct order of the electromagnetic waves from *longer wavelength to shorter wavelength*?



radio waves, infrared, x-rays, microwaves, ultraviolet, visible, gamma rays



radio waves, ultraviolet, x-rays, microwaves, infrared, visible, gamma rays



radio waves, microwaves, visible, x-rays, infrared, ultraviolet, gamma rays



radio waves, infrared, microwaves, ultraviolet, visible, x-rays, gamma rays



radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays



Question 24 1 pts

The voltage across the input terminals of a transformer is 120 V. The primary has 25 loops and the secondary has 50 loops. The voltage the transformer delivers is



60V



120V



240V



30V



Question 25 1 pts

Also review

-> If $P = 10\text{W}$. Whats the energy used in 2 min?

($E = P t$ with t time elapsed in seconds. Power is energy per second)

-> review magnetic force acting on a piece of current carrying piece of wire L placed in B

-> review magnetic force acting on a charge moving at a velocity V

-> review definition of magnetic flux

-> review electromagnetic spectrum.



what...



not ok



mmmmm



ok

Not saved

Submit Quiz